

CHANCELLOR COLLEGE

Mathematical Sciences Department MAT 221-CALCULUS II: TEST 1

Instruction Answer **ALL** Questions **Time Allowed:** 1:30Hrs **Date:** 05/06/2014

1. (a) Find the parametric equations for the line that passes through points $P = (-2, 0, 3)$ and $Q = (3, 5, -2)$. [3 marks]
(b) Deduce the corresponding symmetric equations for the line in 1(a) above. [2 marks]
(c) At what point does the line in 1(a) or 1(b) intersect with the $x - z$ plane? [2 marks]
(d) Find the parametric equation of the plane through $P = (1, 1, -1)$, $Q = (2, 0, 2)$ and $R = (0, -2, 1)$. [5 marks]

2. (a) Differentiate the following functions;
(i.) $f(t) = (t + 1)^{\sqrt{t+1}}$. [5 marks]
(ii.) $g(u) = 4u \sinh(3u^2 - 1)$. [4 marks]
(b) Show that $\int \frac{dv}{\sqrt{b^2 - v^2}} = \sin^{-1}\left(\frac{v}{b}\right) + C$. [5 marks]
(c) Use 2(b) to evaluate $\int \frac{dx}{\sqrt{7 - 6x - x^2}}$. [3 marks]

3. Evaluate;
(a) $\tanh\left(\frac{1}{2} \ln x\right)$. [4 marks]
(b) $\int e^{-x} \sinh(x) dx$. [3 marks]

4. (a) Convert the cartesian equation $y = \frac{x}{x+1}$ into polar equation. [3 marks]
(b) Hence, show that if $r = f(\theta)$, then the length of an arc from $\theta = \theta_1$ to $\theta = \theta_2$ is

$$L = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta.$$

- (c) Use 4(ii.) to find the length of an arc $r = e^\theta$ from $\theta = 0$ to $\theta = 4$. [7 marks]
[4 marks]

5. Sketch and provide a name of;
(a) $r = 1 - \sin \theta$. [5 marks]
(b) $r = 4 \cos 3\theta$. [5 marks]

End of Questions